

City University of Hong Kong
Course Syllabus

offered by Department of Information Systems
with effect from Semester A 2017 / 2018

Part I Course Overview

Course Title: Data Structure and Algorithms

Course Code: IS2239

Course Duration: One Semester (13 weeks)

Credit Units: 3

Level: B2

Proposed Area:
(for GE courses only)

☐ Arts and Humanities

☐ Study of Societies, Social and Business Organisations

☐ Science and Technology

Medium of Instruction: English

Medium of Assessment: English

Prerequisites:
(Course Code and Title) Nil

Precursors:
(Course Code and Title) Nil

Equivalent Courses:
(Course Code and Title) Nil

Exclusive Courses:
(Course Code and Title) Nil

Part II Course Details

1. Abstract

(A 150-word description about the course)

The design and analysis of efficient data structures has long been recognized as a key component for information systems. It is becoming even more important in the big data era as modern information systems may require different data structures to store unstructured data from multiple sources. On completion of this course, students will be able to use the object-oriented paradigm as the framework of choice for the design of data structures. They can design abstract data types (ADT) and associate them with Java interfaces for business applications.

2. Course Intended Learning Outcomes (CILOs)

(CILOs state what the student is expected to be able to do at the end of the course according to a given standard of performance.)

No.	CILOs [#]	Weighting* (if applicable)	Discovery-enriched curriculum related learning outcomes (please tick where appropriate)		
			A1	A2	A3
1.	Describe the concepts object-oriented design for data structures and algorithms.	30%	✓	✓	
2.	Design data structures and algorithms with object-oriented paradigm.	50%	✓	✓	
3.	Evaluate the effectiveness of various data structures for business applications.	20%	✓		✓
		100%			

* If weighting is assigned to CILOs, they should add up to 100%.

[#] Please specify the alignment of CILOs to the Gateway Education Programme Intended Learning outcomes (PILOs) in Section A of Annex.

A1: Attitude

Develop an attitude of discovery/innovation/creativity, as demonstrated by students possessing a strong sense of curiosity, asking questions actively, challenging assumptions or engaging in inquiry together with teachers.

A2: Ability

Develop the ability/skill needed to discover/innovate/create, as demonstrated by students possessing critical thinking skills to assess ideas, acquiring research skills, synthesizing knowledge across disciplines or applying academic knowledge to self-life problems.

A3: Accomplishments

Demonstrate accomplishment of discovery/innovation/creativity through producing /constructing creative works/new artefacts, effective solutions to real-life problems or new processes.

3. Teaching and Learning Activities (TLAs)

(TLAs designed to facilitate students' achievement of the CILOs.)

TLA	Brief Description	CILO No.			Hours/week (if applicable)
		1	2	3	
TLA1. Lecture	Concepts and general knowledge of object-oriented approach to data structures and algorithms.	✓	✓	✓	Lecture: 1 Hour/Week
TLA2. Tutorial	Hands-on exercises of major aspect of object-oriented approach to data structures and algorithms for business problems.		✓	✓	Laboratory: 2 Hours/Week
TLA3. Class Discussion	Questions and answers in lectures and tutorials to get feedback and reflections from students.	✓	✓	✓	

4. Assessment Tasks/Activities (ATs)

(ATs are designed to assess how well the students achieve the CILOs.)

Assessment Tasks/Activities	CILO No.			Weighting *	Remarks [#]
	1	2	3		
Continuous Assessment: 60%					
<u>AT1. Tutorial Exercises & Class Discussion:</u> Each tutorial has in-class exercises to assess students' hands-on skills and understanding of the subject domain.	✓	✓	✓	25%	
<u>AT2. Individual Assignment:</u> The project, including written report and presentation, is required of the students to apply the technical skill sets in the practical analysis and design of solutions and to apply the communication skills in the problem solving with the stakeholders.	✓	✓	✓	15%	
<u>AT3. Mid-term Test:</u> The test serves the purpose of a milestone to test students' ability in the key domain areas and as an indicator of how well the students have performed.	✓	✓	✓	20%	
Examination: 40% (duration: one 2-hour exam)					
<u>AT4. Final Examination:</u> The examination is to assess students' overall competence level in the domain areas.	✓	✓	✓	40%	
* The weightings should add up to 100%.				100%	

* The weightings should add up to 100%.

[#] Remark: Students must pass BOTH coursework and examination in order to get an overall pass in this course.

5. Assessment Rubrics

(Grading of student achievements is based on student performance in assessment tasks/activities with the following rubrics.)

Assessment Task	Criterion	Excellent (A+, A, A-)	Good (B+, B, B-)	Fair (C+, C, C-)	Marginal (D)	Failure (F)
AT1. Tutorial Exercises & Class Discussion	Ability to describe the concepts and design of data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Ability to design data structures and algorithms using object-oriented approach.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Capability to evaluate the data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
AT2. Individual Assignment	Ability to describe the concepts and design of data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Ability to design data structures and algorithms using object-oriented approach.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Capability to evaluate the data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
AT3. Mid-term Test	Ability to describe the concepts and design of data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Ability to design data structures and algorithms using object-oriented approach.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Capability to evaluate the data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
AT4. Final Examination	Ability to describe the concepts and design of data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Ability to design data structures and algorithms using object-oriented approach.	High	Significant	Moderate	Basic	Not even reaching marginal levels
	Capability to evaluate the data structures and algorithms in business problem solving.	High	Significant	Moderate	Basic	Not even reaching marginal levels

Part III Other Information (more details can be provided separately in the teaching plan)

1. Keyword Syllabus

(An indication of the key topics of the course.)

Java Basics

Classes and objects; Strings, wrappers, arrays; Expressions, control flow; Input and output.

Object-oriented Design

Goals, principles and patterns; Inheritance; Interfaces and abstract classes.

Data Structures and Algorithms

Arrays; Linked lists; Stacks; Queues; Trees; Graph algorithms; Sorting; Searching; Recursion and backtracking.

2. Reading List

2.1 Compulsory Readings

(Compulsory readings can include books, book chapters, or journal/magazine articles. There are also collections of e-books, e-journals available from the CityU Library.)

1.	Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser, 2014, <u>Data Structures and Algorithms in Java</u> , 6th edition, ISBN-13:978-1118771334, Wiley.
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2.2 Additional Readings

(Additional references for students to learn to expand their knowledge about the subject.)

1.	Narasimha Karumanchi, 2011, <u>Data Structures and Algorithms Made Easy in Java: Data Structure and Algorithmic Puzzles</u> , 2 nd edition. Cereer Monk Publications.
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